

Calculating Inclinations of Three Stellar Populations Key to Exoplanet Surveys

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Abstract

Stellar inclinations are necessary to characterize exoplanetary systems in three dimensions, primarily through the role of inclinations in calculating the obliquity in the system. Stellar inclinations may furthermore identify the rate of spin-orbit alignment in multistar systems, as well as identify systems well-suited to certain exoplanet detection strategies. Despite the significance of stellar inclinations, the quantity is difficult to measure as it is only inferred from period, radius, and rotational velocity measurements; inclinations are therefore left unresolved for many stars. This proposed work seeks to measure the inclinations of three stellar populations previously highlighted as fruitful exoplanet candidates or hosts: edge-on binaries, targets listed by the NASA Habitable Worlds Observatory (HWO), and exoplanet systems that may demonstrate a large Rossiter-McLaughlin (RM) effect. Using archival photometric and spectroscopic data, open-source programs, and high-performance computing, we will calculate inclination measurements for these three populations. We expect edge-on binaries and RM candidates to have edge-on stellar inclinations, while HWO targets are expected to be pole-on.

Introduction

Uncovering the orbital demographics of exoplanetary systems is of vital importance to contextualize our solar system among our galactic neighborhood. The architecture of exoplanet systems is defined relative to the orientation of the host star (or stars, in the case of multistar systems). One defining geometric feature is the *stellar inclination*, i , the angle between a star's rotational axis and the line-of-sight of a viewer from Earth (*Figure 1*). Stellar rotation was initially theorized to align with the orbit of a protoplanetary disk (Hale 1994); recent work

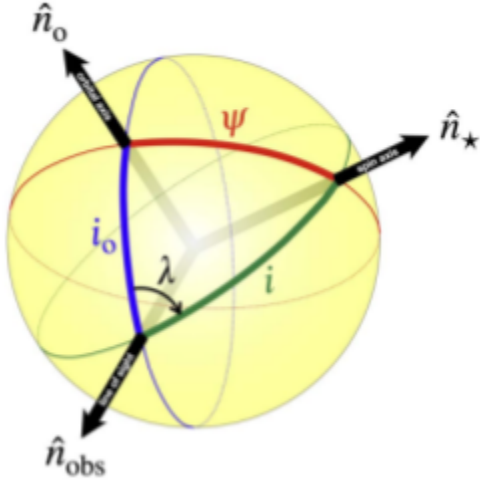


Figure 1. Angular parameters of an exoplanetary system. The stellar spin axis is at an angle i from the line of sight to Earth. From, i and λ , the projected spin-orbit angle derived from RM measurements, ψ , the spin-orbit angle, can be determined (Albrecht et al. 2022).

indicates that this alignment holds true in many, but not all, stellar and planetary populations with a measured spin-orbit angle ψ , referred to as the *obliquity* (*Figure 1*, Albrecht et al. 2022).

Inclination measurements are additionally critical to select top targets for follow-up observations in upcoming surveys. In this application, we assume spin-orbit alignment, where $\psi \sim 0^\circ$ and $i \sim i_0$, the angle of the planets' orbital plane (*Figure 1*). Edge-on stars ($i \sim i_0 \sim 90^\circ$) are well-suited for high-precision radial velocity (RV) spectroscopy—detecting a planet via the gravitational pull on its host—as well as identifying exoplanets transiting across the star. Pole-on systems ($i \sim i_0 \sim 0^\circ$; $i \sim i_0 \sim 180^\circ$)

are ideal for directly imaging planets, as they are never blocked by the star in the line of sight.

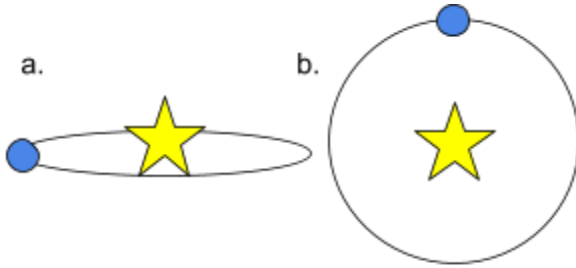


Figure 2. (a) An edge-on system ($i_0 \sim 90^\circ$), viewed from Earth. At the apparent extrema, the planet's gravity pulls the star a detectable amount. (b) A pole-on system ($i_0 \sim 0^\circ$; $i_0 \sim 180^\circ$) as viewed from Earth. The planet is never blocked behind the star.

Despite its significance, the stellar inclinations of exoplanet host and candidate stars are poorly constrained. This is primarily because the stellar inclination cannot be directly measured. Rather, it is inferred from three measurable stellar parameters: the radius R , the rotational period P , and the line-of-sight projection of the rotational velocity $v \sin i$, where i is

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the inclination angle. R is often derived from the spectral energy distribution (e.g. Vines & Jenkins 2022) or the Stefan-Boltzmann law (e.g. Mann et al. 2015). P is taken as the strongest periodicity in a photometric light curve (e.g. McQuillan et al. 2014). The projected rotational velocity can be found by the degree of broadening of absorption lines in the stellar spectrum (e.g. Głęboki & Gnaciński 2005). i is derived by measurements of all three values, via the relation:

$$i = \sin^{-1}\left(\frac{v \sin i}{v}\right) = \sin^{-1}\left(\frac{v \sin i}{2\pi R/P}\right). \quad (1)$$

The proposed work seeks to recover the inclination of three stellar populations of interest to exoplanet searches. (1) 591 bright edge-on binary star systems (1182 stars total; Hand et al. 2025). In these systems, i would demonstrate the rate at which stellar rotation is aligned with binary stars' co-orbital plane. Systems aligned in this way would be prime targets for transit or RV follow-up observations to discover whether this alignment extends to planetary companions.

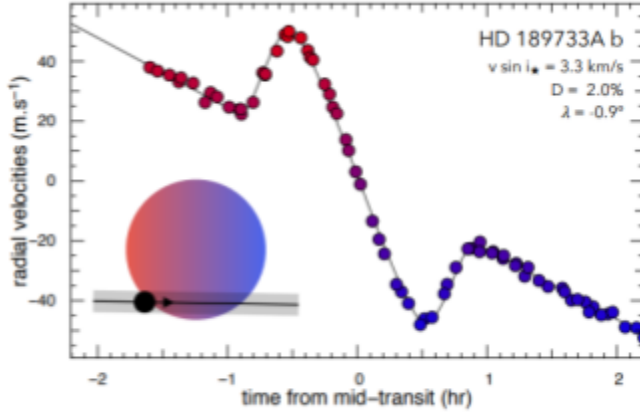


Figure 3. Example of the Rossiter-McLaughlin (RM) effect during the transit of exoplanet HD 189733A b. The star spins clockwise such that the red hemisphere rotates away from the viewer (negative radial velocity), while the blue hemisphere rotates towards (positive radial velocity). When the planet transits, it first covers the red hemisphere and the radial velocity measurement is dominated by positive (blue) values. When the planet blocks the blue hemisphere, the opposite occurs. The colored plot indicates the net radial velocity while that hemisphere is blocked (Triaud 2018).

spinning surface (Figure 3). With a high-precision RM observation (which calculates λ , Figure 1) and a measurement for i , one may derive the spin-orbit angle ψ (Figure 1) and thus better understand the rate of spin-orbit alignment in exoplanetary systems.

(2) 164 bright, nearby stars listed by the NASA Habitable Worlds Observatory (HWO; Mamajek & Stapelfeldt 2023). Derivation of i would identify which systems are pole-on and therefore best suited for HWO's planned direct imaging campaign. (3) Host stars with predicted Rossiter-McLaughlin (RM) semi-amplitudes $> 5 \text{ ms}^{-1}$, initially filtered to all hosts with calculable RM semi-amplitudes via the Exoplanet Follow-up Observing Program (ExoFOP; 7,364 stars maximum; NExSci 2022). The RM effect describes the apparent change in a star's velocity due to a planet transit blocking a portion of the

Hypothesis: I have independent hypotheses for each target population.

- (1) Edge-on binaries are expected to have $i \sim 90^\circ$, assuming that stellar rotations are aligned with the stars' co-orbital plane.
- (2) HWO targets ideally have $i \sim 0^\circ$ or $\sim 180^\circ$, as HWO prioritizes direct imaging. However, the HWO target list was not vetted based on i ; it is possible that there is no correlation.
- (3) The RM candidate population is drawn from the catalog of stars with transiting planets ($i_0 \sim 90^\circ$; *Figures 1 & 2*). I hypothesize that most of these stars will be aligned with their planets' orbital planes, therefore, $i \sim 90^\circ$.

Specific Aim #1: *Infer R, P, and $v \sin i$ from archival data*

While some stars in the aforementioned catalogs have published values of their radii, rotational periods, and line-of-sight projected rotational velocity, there remains a significant population of stars for which these values are poorly constrained or wholly absent. Thus, one aim of this project is to fill as many of these data gaps as possible.

Any missing stellar radii will be found by querying *ARIADNE* (Vines & Jenkins 2022), an automatic pipeline that takes the weighted average of up to six model fits to the spectral energy distribution of a star in order to derive stellar parameters. Due to the size of our sample, *ARIADNE*'s minimal input (the coordinates of the star) is ideal compared to alternatives which require stellar parameters or spectroscopic line list inputs (e.g. Piskunov & Valenti 2017).

Absent period values will be recovered via the *SpinSpotter*¹ code, which provides the capacity to stitch light curve observations together and automatically recovers the rotation period. Data will primarily be drawn from the Transiting Exoplanet Survey Satellite (TESS; Ricker et al. 2014) and Zwicky Transient Facility (ZTF; Bellm et al. 2019) archives, both all-sky survey programs. The high cadence of TESS pointings (30-minute 2018-20, 10-minute 2020-22, 200-second 2022-present) is well-suited to cover short-period rotators; however, each observation period is limited to ~ 30 days, after which the satellite moves to a new field of view. This limit, as well as a mid-observation data gap, restricts resolved TESS periods to ~ 13 days,

¹ <https://github.com/rae-holcomb/SpinSpotter>

though there are notable advancements in extending that baseline (Claytor et al. 2024, Hattori et al. 2025). In contrast, the ground-based ZTF has performed an all-sky survey once every two days for the past six years, allowing the facility to capture slow rotators, albeit at a lower cadence.

Missing rotational velocity measurements will be extracted from archival data taken by the High Resolution Echelle Spectrograph (HIRES; Vogt et al. 1994), the Keck Planet Finder (KPF; Gibson et al. 2016), and the NN-explore Exoplanet Investigations with Doppler spectroscopy (NEID; Schwab et al. 2016), all high precision spectrographs. I will use the *PySME* (Wehrhahn et al. 2022) code, which takes a spectral observation as input and outputs values for the stellar parameters, including $v \sin i$. Like *ARIADNE*, *PySME* was selected due to its minimal user input.

Specific Aim #2: Independent derivation of i

Leveraging measurements present in the literature and those that I derive in *Specific Aim #1*, I will employ the methodology described in Masuda & Winn (2020) to obtain measurements of i with uncertainties. This method (hereafter, M&W20) computes the marginal likelihood and posterior probability of $\cos i$ from likelihood functions for v and $v \sin i$, where $v = 2\pi R/P$. This method is superior to other formulations found in the literature as it recognizes v and $v \sin i$ as related variables, where other strategies, such as quadrature error propagation or previous forms of Bayesian analysis, assume independent uncertainties for the two variables. Thus, the i of many stellar systems has yet to be calculated via M&W20.

Potential Pitfalls and Alternate Strategies:

The variability shown in stellar light curves is due to circulating star spots, causing the star to brighten and darken periodically. However, not all stars exhibit this activity, and therefore their orbital periods cannot be derived. The exact proportion of stars with resolvable rotations is unclear; e.g. with a vetted stellar population observed by the Kepler satellite (Borucki et al. 2010), Reinhold et al. (2013) has resolved rotation periods for 59.3% (24124/40661) while McQuillan et al. (2014) reports rotation periods for 25.6% (34030/133030) of their sample. Thus, it is likely that we will not be able to derive the rotational period, and therefore inclination, of a significant

fraction of stars, which will be discarded from our analysis. Nevertheless, inclinations for even a small fraction of our sample would provide invaluable context to those systems of interest for future surveys and missions.

I also note that M&W20 comes with two assumptions: (1) the data from which v (i.e., R and P , recall *Equation 1*) is derived is independent from the dataset from which $v \sin i$ was derived and (2) v and i are independent variables (contrasted with the faulty assumption that v and $v \sin i$ are independent). While (1) simply requires care in the input data, (2) is troublesome because, as noted in M&W20, no physical laws guarantee v and i independence, e.g. there may be populations of stars with similar v and i , thus correlating the two. Regardless, M&W20 demonstrate that the faulty assumption of independence between v and $v \sin i$ is significantly more inaccurate than the faulty assumption that v and i are independent. As such, I will take care that v and $v \sin i$ are calculated with separate observations, and note any v - i trends I find in the stellar populations as a source of degeneracy.

The sample size of our target populations poses an additional challenge. Whereas analyses akin to Bowler et al. (2023), with a sample size of 62 stars, are able to single out certain stars for further analysis, our sample sizes of the edge-on binaries (1182), HWO targets (164), and RM candidates (7,364) are far too large for this level of inspection. As such, after validating the results of a custom pipeline that performs *Specific Aims #1* and *#2* with a sample of stars with recorded parameters, the majority of the analysis will be run on the high-performance computing network *Bouchet*.

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